

Mahalingam R

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Skills

Languages: TypeScript, JavaScript, Python, C#, SQL, C/C++

Backend & Frameworks: NestJS, Node.js, FastAPI, ASP.NET Core MVC, Celery, Prisma ORM, SQLAlchemy

Databases & Caching: PostgreSQL, pgvector, Redis, Redis Streams, ElastiCache

Cloud & DevOps: AWS (Lambda, ECS Fargate, ECR, S3, SES, CloudFront, ALB, Route 53, ACM, Secrets Manager, EventBridge), GCP (Storage Transfer Service, Cloud Storage), Terraform, Docker, GitHub Actions, CI/CD

AI Engineering: OpenAI GPT-4o, Gemini 2.0 Flash, Anthropic Claude, Whisper, Prompt Engineering, RAG, pgvector semantic search, BM25, Cross-Encoder Re-ranking, RAGAS, sentence-transformers, fastembed

Frontend Technologies: React, Next.js, TypeScript, Tailwind CSS, shadcn/ui, Material UI

Auth & Security: Auth0, JWT, RBAC, RSA-signed cookies, OAuth2

Integrations & Tools: Twilio (WhatsApp), Cashfree, Zoho Books, Stripe, SAP Integration Suite, PDFKit, Sharp, FFmpeg

Work Experience

Ekko Medical Solutions Pvt Ltd, Chennai

Oct 2025 - Present

Software Developer (Backend, AI & Distributed Systems)

- Architected and deployed a real-time transcription microservice using **WebSockets**, **Redis**, and **PostgreSQL**, with FFmpeg, speaker diarization, live buffering, and persistent transcription workflows.
- Designed a **provider-agnostic LLM service** with per-tenant primary/fallback configuration and **automatic failover** across OpenAI, Gemini, and Anthropic; built schema-validated pipelines that convert unstructured input into structured records.
- Engineered a multi-format data extraction pipeline for images, PDFs, and audio with **parallel processing** and cross-category deduplication across 50+ vital types; added **Sharp** image compression to reduce payload size before Vision API calls.
- Provisioned AWS infrastructure end-to-end using **Terraform** and migrated production services to the ap-south-1 (Mumbai) region; deployed containerized services on **ECS Fargate** behind **ALB** with Route 53 and ACM.
- Built **GitHub Actions CI/CD** pipelines for automated ECR deployments and zero-downtime ECS rollouts; implemented cross-cloud **disaster recovery** using scheduled Lambda PostgreSQL backups, S3, and GCP Storage Transfer.
- Implemented **Auth0 JWT** authentication with granular **RBAC** and secure content delivery using **CloudFront**, RSA-signed cookies, S3 Origin Access Control, Secrets Manager, and tenant-scoped storage.
- Developed a subscription billing system integrating **Cashfree** (webhooks, reconciliation), GST and proration logic, Zoho Books invoicing, and **Celery**-based license-lifecycle management.
- Delivered secure multi-channel report distribution over **WhatsApp (Twilio)** and **email (SES)** with PDFKit generation, encrypted S3 storage, and JWT-secured expiring links with OTP fallback.

ROX Hi-TECH, Chennai

June 2024 - Oct 2025

Full Stack Developer

- Built SAP integrations using **SAP Integration Suite** — designed iFlows, managed APIs, and integrated **SAP S/4HANA** with internal applications (earned **SAP Certified Development Associate** certification).
- Developed full-stack HR data management application using **ASP.NET Core 8 (MVC)** with Razor views and **PostgreSQL** — implemented file upload forms, request-tracking workflow with auto-generated request numbers, and admin approve/reject functionality.
- Integrated SAP systems via **ODATA APIs** to fetch HR master data (payroll areas, personnel areas, gender data) through ASP.NET Core backend services, surfacing the data into MVC views.
- Built admin dashboard module enabling master data management — infotype creation, role-based admin assignment, and end-to-end request approval/rejection workflow.
- Used **AJAX with jQuery** to send asynchronous POST requests from Razor views to MVC controller actions, enabling seamless form submissions without full page reloads.

Project Work

- **Full-Stack E-Commerce Platform (Shoppy)**

Live Demo

Built a production-grade e-commerce platform with Next.js 14 (App Router, Server Components, Server Actions) and NestJS backend deployed on AWS. Implemented JWT authentication with HttpOnly cookies and role-based access control (Admin/User), Stripe payment integration with webhook-driven order management, and product image uploads to AWS S3. Enabled real-time product updates via Socket.io WebSocket gateway. Set up CI/CD pipeline using AWS CodePipeline and CodeBuild for automated deployments to Elastic Beanstalk, with CloudFront CDN for edge caching and HTTPS termination. Designed a responsive UI with Tailwind CSS and Material UI.

Tech Stack: Next.js 14, NestJS, TypeScript, PostgreSQL, Prisma ORM, Stripe, AWS (Elastic Beanstalk, RDS, S3, CloudFront, CodePipeline, CodeBuild), Socket.io, Tailwind CSS, Material UI

- **MedRAG Assistant — Medical RAG Knowledge System**

Live Demo

Built a production-grade Retrieval-Augmented Generation (RAG) system for medical document Q&A using FastAPI and React. Implemented hybrid retrieval (BM25 + semantic search with pgvector) merged via Reciprocal Rank Fusion, cross-encoder re-ranking (ms-marco-MiniLM) to improve precision from top-20 candidates to top-5, and query rewriting for multi-turn conversations. Integrated Groq LLaMA 3.3 70B for grounded answer generation with source citations and SSE streaming. Built a RAGAS evaluation pipeline (OpenAI gpt-4o-mini judge) achieving 0.93 context recall and 0.86 faithfulness across 15 hand-written medical Q&A pairs. Deployed backend on Render, frontend on Vercel, PostgreSQL on Neon.

Tech Stack: FastAPI, Python, React, TypeScript, PostgreSQL, pgvector, fastembed (ONNX), Groq LLaMA 3.3 70B, RAGAS, Tailwind CSS, shadcn/ui

Education

SASTRA Deemed University

B.Tech. in Computer Science and Business Systems

Aug 2020 - Jun 2024

CGPA: 7.2/10

Certifications

- **SAP Certified Development Associate – SAP Integration Suite** — Validated skills in designing, developing, and managing integrations using SAP Integration Suite.